ANALYSTLAB AFRICA

Data Science Internship Programme

Heart Disease Prediction
Project Continuity Summary

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

Prepared for the AnalystLab Africa Data Science Internship Programme
Week 3 Submission – Advanced Data Analysis, Statistical Testing & Feature Engineering

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1. Project Transition Note

The Week 1 submission focused on an Employee Attrition Analysis project for ABC Manufacturing Ltd. The objective was to investigate employee characteristics and workplace factors associated with employee turnover. The target variable in that project was employee attrition.

From Week 2, the project scope changed to the Heart Disease Prediction dataset. Therefore, Week 3 continues directly from the Heart Disease Prediction analysis completed in Week 2, rather than treating the Week 1 employee attrition dataset and the Week 2 heart disease dataset as one continuous analytical dataset.

The Week 1 project nevertheless provides useful experience in business understanding, exploratory analysis and data interpretation. The Heart Disease Prediction project from Week 2 is the direct analytical foundation for Week 3.

2. Business Problem

Heart disease is an important public-health problem, and cardiovascular disease continues to represent a major burden globally. The American Heart Association's 2026 statistics update highlights the importance of cardiovascular risk factors including blood pressure, cholesterol, glucose control, obesity and other health-related factors (American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Committee, 2026).

Within this project, the business and analytical problem is to investigate whether demographic, symptom and clinical characteristics contained in the dataset can be used to identify patterns associated with the presence of heart disease.

The Week 2 assignment defined the project as a Heart Disease Prediction problem, with the objective of preparing the dataset for predicting whether a patient is likely to have heart disease.

The ultimate purpose is to establish a sufficiently robust analytical foundation for predictive modelling. A future predictive model could potentially help identify patient profiles associated with a higher likelihood of heart disease and support further assessment.

However, the current project should not be interpreted as a clinically validated diagnostic system. The analysis is intended for data-science and predictive-modelling purposes, and any real-world clinical application would require appropriate validation and professional oversight (American Heart Association, 2025).

3. Project Objective

The overall objective of the project is to investigate the characteristics and relationships associated with heart disease and prepare a refined dataset suitable for predictive modelling.

Week 2 primarily addressed data preparation and preprocessing. This included identifying data-quality problems, correcting invalid values, engineering initial features, treating outliers, encoding categorical variables and scaling selected numerical variables.

Week 3 extends this work by moving from basic exploratory analysis towards evidence-based analytical investigation. The Week 3 objectives are to:

1. Conduct advanced exploratory data analysis.
2. Investigate distributions, variability and unusual patterns.
3. Examine important bivariate and multivariate relationships.
4. Validate relevant statistical assumptions.

5. Conduct at least four appropriate statistical analyses or hypothesis tests.
6. Determine whether observed relationships are statistically significant.
7. Engineer at least three meaningful features where justified.
8. Evaluate original and engineered features.
9. Investigate redundancy and potential multicollinearity.
10. Identify potentially useful and less useful predictors.
11. Refine the Week 2 dataset based on analytical evidence.
12. Produce a final modelling dataset for Week 4.
13. Translate analytical findings into business-focused recommendations.

This represents a progression from data preparation → deeper analysis → statistical evidence → feature refinement → predictive modelling.

4. Target Variable

The confirmed target variable is HeartDisease. HeartDisease represents the outcome that future predictive models will attempt to predict.

The remaining variables represent potential explanatory or predictor variables.

The Week 2 preprocessing process produced a clean_encoded dataset containing 918 observations and 20 columns, including the target variable and engineered/encoded features.

The Week 3 analysis will therefore focus on understanding which of these variables, individually or in combination, provide meaningful information about HeartDisease.

5. Summary of Previous Work

5.1 Week 1 Context

The Week 1 project investigated employee attrition at ABC Manufacturing Ltd. The analysis focused on understanding employee characteristics and workplace factors that may be associated with employees leaving an organisation.

This project is separate from the current Heart Disease Prediction project. Therefore, Week 1 findings are not being transferred to the heart disease dataset.

The main continuity between the two weeks is the application of a structured data-science process involving:

- Business understanding
- Data exploration
- Data-quality assessment
- Analytical interpretation
- Evidence-based decision-making

The direct dataset continuity for Week 3 begins with the Heart Disease Prediction work completed in Week 2.

6. Key Findings from Week 2

The Week 2 exploratory analysis identified several potentially important relationships that require deeper investigation.

6.1 Target Distribution

The target variable was relatively balanced, although there were somewhat more observations representing heart disease than no heart disease. Approximately 55% of observations represented heart disease, compared with approximately 45% without heart disease.

This is important for Week 3 because substantial class imbalance does not appear to be a major issue at this stage, although the distribution should still be considered when future predictive models are evaluated.

6.2 Age and Heart Disease

The Week 2 analysis indicated that patients with heart disease tended to be older than patients without heart disease. Age was also negatively related to Maximum Heart Rate and positively related to Oldpeak.

This suggests that Age should be investigated more rigorously during Week 3. A key question is:

Is the observed difference in age between patients with and without heart disease statistically significant?

6.3 Maximum Heart Rate

MaxHR demonstrated a negative association with HeartDisease. The Week 2 analysis observed that patients with heart disease generally had lower maximum heart-rate values. The encoded analysis produced an approximate correlation of -0.40 between MaxHR and the target.

This makes MaxHR an important candidate for hypothesis testing and further feature evaluation.

6.4 Oldpeak

Oldpeak showed a positive relationship with heart disease. Patients with heart disease generally displayed higher Oldpeak values, with an observed correlation of approximately 0.41 with the target.

Week 3 should determine whether this observed relationship is statistically significant and whether Oldpeak provides useful information beyond other related variables.

6.5 ST Slope

The encoded ST_Slope categories showed some of the strongest observed relationships with the target. ST_Slope_Flat had an approximate positive correlation of 0.55, while ST_Slope_Up had an approximate negative correlation of -0.62 with HeartDisease.

These results make ST_Slope one of the highest-priority variables for Week 3 statistical investigation.

Because ST_Slope is categorical, the relationship should be evaluated using an appropriate categorical method such as a Chi-Square test of independence, rather than relying solely on correlations calculated after encoding.

6.6 Exercise Angina

ExerciseAngina_Y showed a positive association with heart disease, with an approximate correlation of 0.49.

This suggests that Exercise Angina should also be investigated using a categorical hypothesis test.

6.7 Chest Pain Type

ChestPainType_ATA showed a negative association with heart disease, with an approximate correlation of -0.40.

Week 3 will investigate the broader relationship between ChestPainType and HeartDisease, rather than focusing only on one encoded category.

6.8 Sex

The Week 2 dataset contained 725 male observations and 193 female observations. Sex_M showed an approximate positive association of 0.31 with the target.

The unequal group sizes should be considered when interpreting this relationship.

Week 3 should determine whether the observed association is statistically significant rather than assuming that correlation alone establishes an important relationship.

7. Major Data-Quality Issues Identified in Week 2

The original dataset contained 918 observations and 12 variables. Initial checks did not identify conventional missing values, but further investigation identified invalid zero values in clinical variables.

7.1 Cholesterol

A total of 172 records had Cholesterol = 0. These observations were treated as invalid rather than genuine cholesterol measurements. The values were converted to missing values and replaced using the median of the valid observations. The resulting median was 237.0.

7.2 RestingBP

One observation contained RestingBP = 0. This was treated as an invalid clinical measurement and replaced using the median of the valid observations. The median was 130.0.

7.3 Duplicate Records

The dataset was checked for duplicate records. The result was 0 duplicate records.

8. Major Preprocessing Actions Completed in Week 2

Week 2 converted the raw dataset into a structured dataset suitable for subsequent analysis. The major actions were:

Invalid-value treatment: Invalid zero values in Cholesterol and RestingBP were replaced through median imputation. The median was chosen over the mean because it is less sensitive to skewed distributions and remaining extreme values, providing a more robust central tendency for imputed clinical measurements (Little & Rubin, 2019).

Duplicate checking: No duplicate observations were identified.

Feature engineering: Two initial engineered variables were created:

- Age_Group
- BP_Category

Outlier treatment: MaxHR and Oldpeak were investigated using boxplots and the IQR method. Potential outliers were capped rather than removed. FastingBS was treated as a binary feature and was therefore not subjected to continuous-variable IQR treatment.

One-Hot Encoding: Categorical variables were transformed using One-Hot Encoding. This approach creates binary indicators for categories and is commonly used to represent categorical features numerically for machine-learning algorithms (Potdar et al., 2017; scikit-learn developers, 2026a).

Feature Scaling: MaxHR and Oldpeak were standardised using StandardScaler. Standardisation centres a feature around its mean and scales it according to its standard deviation (Pedregosa et al., 2011; scikit-learn developers, 2026b).

Final Dataset: The resulting clean_encoded dataset contained 918 observations and 20 columns and was prepared for subsequent machine-learning analysis.

9. Why Week 3 Requires Deeper Investigation

The Week 2 analysis identified correlations and visual patterns, but these findings should not automatically be interpreted as statistically significant relationships.

For example, a correlation between MaxHR and HeartDisease indicates an observed association within the sample. It does not establish that the relationship is statistically significant or that MaxHR causes heart disease.

Week 3 therefore introduces statistical testing and assumption validation to answer questions such as:

- Are the observed group differences statistically significant?
- Are relationships strong enough to be analytically useful?
- Are the assumptions of the selected statistical tests satisfied?
- Do engineered variables add useful information?
- Are some variables redundant?
- Are any variables affected by multicollinearity?
- Are there potential data-leakage concerns?
- Which features should be retained for Week 4?

10. Key Questions for Week 3

Based on the findings from Week 2, the following research questions will guide the advanced analysis.

Research Question 1

Is Age significantly different between patients with and without heart disease?

This will investigate whether the observed age difference is supported by statistical evidence.

Research Question 2

Is Maximum Heart Rate significantly different between patients with and without heart disease?

This follows from the negative association identified in Week 2.

Research Question 3

Is Oldpeak significantly different between patients with and without heart disease?

This follows from the positive relationship identified during exploratory analysis.

Research Question 4

Is ST Slope statistically associated with heart disease?

This is particularly important because ST_Slope produced some of the strongest associations in the Week 2 analysis.

Research Question 5

Is Exercise Angina statistically associated with heart disease?

The Week 2 relationship makes this another important categorical variable for formal hypothesis testing.

Research Question 6

Do Age_Group and BP_Category provide additional analytical value?

These engineered features should not automatically be retained simply because they were created in Week 2. Week 3 will evaluate whether they provide useful information beyond the original numerical variables.

11. Week 3 Advanced Exploratory Direction

The Week 3 EDA will move beyond repeating Week 1/Week 2 descriptive analysis. The analysis will investigate:

Numerical variables

The analysis will investigate:

- Distribution
- Skewness
- Variability
- Potential unusual observations
- Differences by heart disease status
- Relationships between important numerical variables

Potential variables include: Age, RestingBP, Cholesterol, MaxHR and Oldpeak.

Categorical variables

The analysis will investigate:

- Frequency
- Percentage distribution
- Group differences
- Relationship with HeartDisease

Important variables include: Sex, ChestPainType, FastingBS, RestingECG, ExerciseAngina, ST_Slope, Age_Group and BP_Category.

Bivariate relationships

The analysis will examine:

- Age vs MaxHR
- Age vs Oldpeak
- MaxHR vs Oldpeak
- RestingBP vs Cholesterol
- Numerical variables vs HeartDisease
- Categorical variables vs HeartDisease

Multivariate relationships

The analysis will examine combinations of variables rather than isolated relationships. For example:

- Age + MaxHR + HeartDisease
- Age + Oldpeak + HeartDisease
- ST_Slope + ExerciseAngina + HeartDisease
- Age_Group + BP_Category + HeartDisease

This is important because heart disease-related patterns may involve multiple characteristics simultaneously.

12. Statistical Investigation

Week 3 will include at least four statistical analyses. The planned analyses include:

Research Question	Potential Method
Age vs HeartDisease	Independent Samples t-test / Mann–Whitney U
MaxHR vs HeartDisease	Independent Samples t-test / Mann–Whitney U
Oldpeak vs HeartDisease	Independent Samples t-test / Mann–Whitney U
ST_Slope vs HeartDisease	Chi-Square Test
ExerciseAngina vs HeartDisease	Chi-Square Test

The final choice between parametric and non-parametric tests will be based on the distribution and assumptions observed in the Week 3 analysis.

For each test, the notebook will document:

1. Research question
2. Objective
3. Null hypothesis
4. Alternative hypothesis
5. Statistical method
6. Methodological justification
7. Test statistic
8. P-value
9. Decision
10. Interpretation
11. Business implication

Importantly, actual statistical results will only be reported after the tests have been run on the Week 2 dataset.

13. Advanced Feature Engineering

Week 3 requires at least three meaningful engineered features where appropriate. The existing Week 2 features provide a starting point:

Feature 1 — Age_Group

Age is transformed into meaningful age categories.

Purpose: To investigate whether heart disease patterns differ across broad age groups.

Potential value: It may capture non-linear age-related patterns that are less obvious when Age is treated only as a continuous variable.

Feature 2 — BP_Category

RestingBP is transformed into project-defined categories.

Purpose: To provide a categorical representation of blood-pressure levels.

Potential value: It can facilitate group comparisons and categorical statistical analysis.

These categories should be described as project-defined analytical categories, rather than presented as a clinical diagnostic classification.

Feature 3 — Heart_Rate_Age_Ratio

A candidate ratio feature can be created as:

Heart_Rate_Age_Ratio = MaxHR / Age

Purpose: To represent Maximum Heart Rate relative to patient age.

Potential value: The ratio may provide a different representation of the relationship between age and maximum heart rate than either variable individually.

However, its usefulness must be tested rather than assumed.

Additional candidate feature — ST_Exercise_Risk

An interaction between ST_Slope and ExerciseAngina may also be investigated.

Purpose: To capture the combined pattern of two variables that showed potentially important relationships with heart disease.

Potential value: The interaction could identify combinations of characteristics that are more informative than either variable considered independently.

Again, the feature should be retained only if analytical evidence supports its usefulness.

14. Feature Evaluation and Selection

Week 3 will evaluate both the original and newly engineered variables. The evaluation will consider:

- Correlation
- Statistical significance
- Effect size
- Feature relationships
- Multicollinearity
- Redundancy
- Potential data leakage
- Domain relevance
- Potential modelling value

A feature will **not** be removed simply because it is correlated with another feature.

Instead, each major decision will be documented using: *Feature → Method → Evidence → Decision → Justification*

For example:

Feature	Evidence	Potential Decision
Age	Correlation + hypothesis test	Retain/remove based on evidence
MaxHR	Correlation + hypothesis test	Retain/remove based on evidence
Oldpeak	Correlation + hypothesis test	Retain/remove based on evidence
ST_Slope	Chi-square + group analysis	Retain/remove based on evidence
Age_Group	Chi-square + comparison with Age	Evaluate
BP_Category	Chi-square + comparison with RestingBP	Evaluate
Heart_Rate_Age_Ratio	Correlation/feature evaluation	Evaluate
ST_Exercise_Risk	Association/feature evaluation	Evaluate

The final decisions will be based on the actual Week 3 analytical results.

15. Dataset Refinement

The Week 3 dataset will build upon the Week 2 dataset rather than repeating the complete cleaning process. The refinement process will be:

- Week 2 clean_encoded dataset
- Advanced EDA
- Statistical testing
- Feature engineering
- Feature evaluation
- Feature retention/removal decisions
- Final Week 3 modelling dataset

The final dataset will become the foundation for Week 4 Machine Learning.

16. Expected Analytical Contribution of Week 3

The major contribution of Week 3 will be moving from:

"These variables appear to be related to heart disease."

towards:

"These relationships have been investigated statistically, their assumptions have been assessed, their practical relevance has been evaluated, and the evidence supports retaining or removing these features before modelling."

This distinction is important because exploratory relationships alone do not provide sufficient evidence for final feature-selection decisions.

17. Limitations to Carry Forward

Several limitations should be considered during Week 3 and Week 4.

Dataset size: The dataset contains 918 observations. Although this is sufficient for the planned exploratory and statistical analysis, it remains a relatively modest dataset for generalising conclusions to broader populations.

Invalid values: The presence of 172 zero Cholesterol values and one zero RestingBP value indicates limitations in the original data collection or representation. Median imputation provides a practical preprocessing solution but does not recover the original patient measurements (Little & Rubin, 2019).

Demographic imbalance: The dataset contains considerably more male than female observations, which may affect the interpretation and generalisability of relationships involving Sex.

Observational data: Observed associations should not automatically be interpreted as causal relationships.

Clinical generalisation: The dataset should not be assumed to represent all populations or healthcare settings. Further external validation would be necessary before applying a predictive model in a real clinical environment.

18. Week 3 Continuity Statement

The Week 2 project established a cleaned and machine-learning-ready Heart Disease Prediction dataset. The analysis identified potentially important relationships involving Age, Maximum Heart Rate, Oldpeak, ST Slope, Exercise Angina, Chest Pain Type and other patient characteristics.

Week 3 will build directly upon these findings by conducting advanced exploratory analysis and formal statistical investigation.

The progression can therefore be summarised as:

Week 2 established data quality and preprocessing; Week 3 will establish analytical and statistical evidence; Week 4 will use the refined evidence-based dataset for predictive modelling.

This creates a clear and logical progression across the project:

Data Preparation → Advanced Analysis → Statistical Validation → Feature Refinement → Machine Learning

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